NVIDIA ACADEMY

NVIDIA AI Software Ecosystem

Unit 5 - Summary

- **NVIDIA AI Software:** NVIDIA AI Software Ecosystem includes vGPU Foundation, Frameworks for Deep Learning, and an Overview of the Software Stack.
- **NVIDIA AI Enterprise**: NVIDIA AI Enterprise is an end-to-end AI software suite that includes over 50 frameworks and pre-trained models.
- **NGC Catalog:** NGC Catalog is a GPU-optimized hub for AI and HPC software, including containers, pre-trained models, and Helm charts.
- **AI Workflows:** NVIDIA AI Workflows are prepackaged reference applications that accelerate the path to AI outcomes, reducing cost and improving accuracy.

What is vGPU?

vGPU stands for Virtual GPU, which is a foundational technology for AI that delivers GPU acceleration to every visual workload.

What are the benefits of GPU virtualization?

The benefits of GPU virtualization include bare-metal performance, insight and tools, business continuity and workload balancing, resource sharing and improved utilization, infrastructure and data security, and operational management.

What are ML/DL frameworks?

Machine learning and deep learning frameworks are building blocks for designing, training, and validating machine learning models and deep neural networks, through a high-level programming interface.

What is the NVIDIA Deep Learning Software Stack?

The NVIDIA Deep Learning Software Stack includes CUDA, NVIDIA Container Runtime, NGC Containers, and DL Frameworks.

What is NVIDIA AI Enterprise?

NVIDIA AI Enterprise is the software layer of the most advanced AI platform, which includes over 50 frameworks and pretrained models.

What are NVIDIA AI Workflows?

NVIDIA AI Workflows are prepackaged reference applications to rapidly automate businesses with AI, reducing the cost of developing and deploying AI solutions.